

A.6 Coordinate diffusion

For our model we perform diffusion on the *fractional* coordinates and outline the approach in the following. See Appendix A.6.3 for a brief outline why we favor fractional coordinate diffusion over Cartesian. The fractional coordinates in a crystal structure live in a Riemannian manifold referred to as the flat torus $\mathbb{T}^3 = \mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{S}^1$, i.e., it is the quotient space $\mathbb{R}^3/\mathbb{Z}^3$ with equivalence relation:

$$\mathbf{x} + \mathbf{k} \sim \mathbf{x}, \quad \mathbf{k} \in \mathbb{Z}^3. \quad (\text{A18})$$

Thus, adding Gaussian noise to fractional coordinates naturally corresponds to sampling from a *wrapped* normal distribution, whose probability density is

$$\mathcal{N}_W(\bar{\mathbf{x}}; \mathbf{x}, \sigma^2 \mathbf{I}, \mathbf{I}) = \sum_{\mathbf{k} \in \mathbb{Z}^3} \mathcal{N}(\bar{\mathbf{x}}; \mathbf{x} - \mathbf{k}, \sigma^2 \mathbf{I}), \quad (\text{A19})$$

where $\mathcal{N}_W(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \mathbf{B})$ denotes a wrapped normal distribution with mean $\boldsymbol{\mu}$, covariance matrix $\boldsymbol{\Sigma}$, and periodic boundaries $\mathbf{B}$. If the periodic boundaries are $[0, 1)^3$, i.e., $\mathbf{B} = \mathbf{I}$, we write $\mathcal{N}_W(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for brevity.

For the diffusion of the atom coordinates we use variance-exploding diffusion, i.e., the variance of the diffusion process increases exponentially with diffusion time. This has the advantage that the prior distribution $q(\mathbf{x}_T)$ is particularly simple, i.e., the uniform distribution in the range $[0, 1)^3$. Jing et al. [67] use this approach for torsional angles—which live in a 1D flat torus—in small molecule generation. The one-shot noising process of the fractional coordinates is therefore defined as

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}_W(\mathbf{x}_t; \mathbf{x}_0, \sigma_t^2 \mathbf{I}). \quad (\text{A20})$$

A.6.1 Variance adjustment for atomic density

One limitation of using a constant variance for the fractional coordinate diffusion is that the diffusion in Cartesian space will have difference variance depending on the size of the unit cell. This limitation becomes clear if we express the distribution of the Cartesian coordinates $\tilde{\mathbf{x}}_t$ using Eq. (A2) via linear transformation of a Gaussian random variable $\mathbf{x}_t$:

$$q(\tilde{\mathbf{x}}_t | \mathbf{x}_0, \mathbf{L}_t) = \mathcal{N}_W(\tilde{\mathbf{x}}_t; \mathbf{L}_t \mathbf{x}_0, \sigma_t^2 \mathbf{L}_t \mathbf{L}_t^\top, \mathbf{L}_t). \quad (\text{A21})$$

Observe that the covariance matrix $\boldsymbol{\Sigma}_t = \sigma_t^2 \mathbf{L}_t \mathbf{L}_t^\top$ of the noisy Cartesian coordinates depends on the lattice. Thus, the (generalized) variance of the noise distribution also depends on the size of the unit cell, i.e., $|\det(\boldsymbol{\Sigma}_t)| = (\sigma_t^3 |\det \mathbf{L}_t|)^2$.

We mitigate this effect by scaling the variance in fractional coordinate diffusion based on the size of the unit cell. Assuming roughly constant atomic density $d(\mathbf{L}_t) = \frac{n}{\text{Vol}(\mathbf{L}_t)} \propto 1 \Leftrightarrow \text{Vol}(\mathbf{L}_t) = |\det \mathbf{L}_t| \propto n$. We therefore choose to scale σ_t accordingly, i.e.,

$$\sigma_t(n) = \frac{\sigma_t}{\sqrt[3]{n}}, \quad (\text{A22})$$

such that $|\det(\Sigma_t)| = \left(\frac{\sigma_t^3}{n} |\det \mathbf{L}_t|\right)^2$ is no longer proportional to n .

A.6.2 Score computation for fractional coordinates

Recall from Eq. (A4) that training diffusion models requires computing the score function for the one-shot transition kernel. However, for the wrapped normal distribution in Eq. (A19), (log-)likelihood and score computation are intractable because of the infinite sum. Given the thin tails of the normal distribution, both can be approximated reasonably well with a truncated sum. More specifically, the score function of the isotropic wrapped normal distribution can be expressed as

$$\nabla_{\bar{\mathbf{x}}} \log q_{\sigma}(\bar{\mathbf{x}}|\mathbf{x}) = \sum_{\mathbf{k} \in \mathbb{Z}^3} w_{\mathbf{k}} \frac{\bar{\mathbf{x}} - \mathbf{x} + \mathbf{k}}{\sigma^2}, \quad (\text{A23})$$

where

$$w_{\mathbf{k}} = \frac{1}{Z} \exp\left(-\frac{\|\bar{\mathbf{x}} - \mathbf{x} + \mathbf{k}\|^2}{2\sigma^2}\right), \quad Z = \sum_{\mathbf{k}' \in \mathbb{Z}^3} \exp\left(-\frac{\|\bar{\mathbf{x}} - \mathbf{x} + \mathbf{k}'\|^2}{2\sigma^2}\right). \quad (\text{A24})$$

A.6.3 Fractional vs Cartesian coordinate diffusion

Instead of diffusing fractional coordinates as in MatterGen, one could diffuse Cartesian coordinates, e.g., as done in CDVAE [4] (and in generative methods for molecules [68]).

However, this approach is not suitable for our framework. To see this, note that while in CDVAE the lattice $\mathbf{L}$ is fixed during the diffusion of the atom coordinates, we diffuse the lattice simultaneously to the atom coordinates (and atomic species). This makes diffusion of Cartesian coordinates dependent on the lattice diffusion because the wrapped normal’s covariance matrix and periodic boundaries at diffusion timestep t depend on knowing the lattice matrix $\mathbf{L}_t$. Consequently, our diffusion process from Eq. (A9) no longer factorizes into lattice and coordinates and needs to be adapted:

$$\begin{aligned} & q(\tilde{\mathbf{X}}_{t+1}, \mathbf{L}_{t+1}, \mathbf{A}_{t+1} | \tilde{\mathbf{X}}_t, \mathbf{L}_t, \mathbf{A}_t) \\ &= q(\tilde{\mathbf{X}}_{t+1} | \tilde{\mathbf{X}}_t, \mathbf{L}_{t+1}, \mathbf{L}_t) q(\mathbf{L}_{t+1} | \mathbf{L}_t) q(\mathbf{A}_{t+1} | \mathbf{A}_t). \end{aligned} \quad (\text{A25})$$

Here, we need to condition $q(\tilde{\mathbf{X}}_{t+1})$ on $\mathbf{L}_{t+1}$ and $\mathbf{L}_t$ because in order to convert the Cartesian coordinates at time step t to time step $t+1$ we first need to convert $\tilde{\mathbf{x}}_t$ to fractional coordinates using $\mathbf{L}_t^{-1}$, and then to Cartesian coordinates at $t+1$ using $\mathbf{L}_{t+1}$.

The one-shot distribution of noisy Cartesian coordinates (similar to Eq. (A20) for the fractional case) becomes:

$$q(\tilde{\mathbf{x}}_t | \tilde{\mathbf{x}}_0, \{\mathbf{L}_{t'}\}_{t'=1}^t) = \mathcal{N}_{\mathbf{W}}\left(\tilde{\mathbf{x}}_t; \mathbf{L}_t \mathbf{L}_0^{-1} \tilde{\mathbf{x}}_0, \mathbf{L}_t \left(\sum_{t'=1}^t \sigma_{t'}^2 \mathbf{L}_{t'}^{-1} (\mathbf{L}_{t'}^{\top})^{-1}\right) \mathbf{L}_t^{\top}, \mathbf{L}_t\right). \quad (\text{A26})$$

Observe that we require the entire trajectory of noisy lattices $\mathbf{L}_1, \dots, \mathbf{L}_t$ in order to express the noise distribution of the Cartesian atomic coordinates. This means that we first need to sample the *entire* diffusion trajectory of the lattice, which is slow. Further, we have found computing the one-shot covariance matrix for the Cartesian coordinates to be numerically unstable for long diffusion trajectories. We therefore use the diffusion process of fractional coordinates described in the previous section.

A.7 Lattice diffusion

In addition to the diffusion of the atom types and coordinates described above, we also diffuse and denoise the lattice $\mathbf{L}$ in our approach. We use variance-preserving diffusion, as variance-exploding diffusion would lead to extremely large unit cells in the noisy limit. Those are challenging to handle for a graph neural network (GNN) with a fixed edge cutoff and would require the model to learn scores over a wide range of different length scales.

A.7.1 Fixed-rotation lattice diffusion

As the distribution of materials is invariant to global rotation, we can either choose a rotation-invariant prior distribution over unit cells, or decide on a canonical rotational alignment that we use throughout diffusion and denoising. We opt for the latter, as it gives us some more flexibility designing the diffusion process. Here, we choose to represent the lattice as a symmetric matrix. We can do so via the polar decomposition based on the singular value decomposition:

$$\mathbf{L} = \mathbf{U}\tilde{\mathbf{L}}, \quad \mathbf{U} = \mathbf{W}\mathbf{V}^\top, \quad \tilde{\mathbf{L}} = \mathbf{V}\mathbf{\Sigma}\mathbf{V}^\top, \quad (\text{A27})$$

where $\mathbf{W}$ and $\mathbf{V}$ are the left and right singular vectors of $\mathbf{L}$, respectively, and $\mathbf{\Sigma}$ is the diagonal matrix of singular values. $\mathbf{U}$ is a rotation matrix and $\tilde{\mathbf{L}}$ is a symmetric positive-definite matrix.

We restrict our entire forward lattice diffusion to symmetric matrices by enforcing the noise $\mathbf{z} \in \mathbb{R}^{3 \times 3}$ on the lattice to be symmetric, e.g., by only modeling the upper-triangular part of the matrix and mirroring it to the lower triangular part. Notice that this effectively fixes the rotation, i.e., we only have six degrees of freedom left. Going forward, we only consider symmetric lattices and lattice noise.

A.7.2 Lattice diffusion with custom limit mean and variance

Naively using variance-preserving diffusion independently on the lattice vectors leads to the following forward diffusion distribution:

$$q(\mathbf{L}_t | \mathbf{L}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{L}_0, (1 - \bar{\alpha}_t) \mathbf{I}). \quad (\text{A28})$$

However, for large t we observed that the majority of the resulting unit cells have very small volume and steep angles, which means that the atoms are extremely densely

packed inside the noisy cells. We therefore modify the diffusion process as follows:

$$q(\mathbf{L}_t|\mathbf{L}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{L}_0 + (1 - \sqrt{\bar{\alpha}_t})\mu(n)\mathbf{I}, (1 - \bar{\alpha}_t)\sigma_t^2(n)\mathbf{I}), \quad (\text{A29})$$

which yields the following limit distribution for $T \rightarrow \infty$:

$$q(\mathbf{L}_T) = \mathcal{N}(\mu(n)\mathbf{I}, \sigma_T^2(n)\mathbf{I}). \quad (\text{A30})$$

Thus, in the limit distribution we have a tendency towards cubic lattices $\mathbf{L} \propto \mathbf{I}$, which often occur in nature and have a relatively narrow range of volumes. Further, the lattice vector angles when sampling from the prior are mostly concentrated between 60° and 120° . This aligns both with the range of angles of Niggli-reduced cells as well as the initialization range of cell vector angles in RSS [69], suggesting that this is a good starting point for the generation process.

To understand the choice of the mean in the limit distribution, recall that the volume of a parallelepiped $\mathbf{L}$ is $|\det \mathbf{L}|$. We introduce a scalar coefficient $\mu(n)$ that depends on the number of atoms in the cell to make the atomic density of the mean noisy lattice roughly constant for differently sized systems. Setting $\mu(n) = \sqrt[3]{nc}$, the volume of the prior mean becomes $\text{Vol}(\sqrt[3]{nc}\mathbf{I}) = nc$. Thus, the atomic density of the prior mean becomes $d(\mu(n)\mathbf{I}) = \frac{n}{\text{Vol}(\mu(n)\mathbf{I})} = \frac{1}{c}$. We can set c to the inverse average density of the dataset as a reasonable prior.

Similarly, we adjust the variance in the limit distribution to be proportional to the volume, such that the signal-to-noise ratio of the noisy lattices is constant across the numbers of nodes. To this end, we set the limit standard deviation as $\sigma(n) = \sqrt[3]{n\nu}$. Thus, for a diagonal entry of the lattice matrix the signal-to-noise-ratio in the limit is $\lim_{t \rightarrow \infty} \frac{|\mu(n)|}{\sigma(n)} = \frac{\sqrt[3]{nc}}{\sqrt[3]{n\nu}} = \left(\frac{c}{\nu}\right)^{1/3}$ and therefore independent of the number of atoms.

A.8 Architecture of the score network

We employ an SE(3)-equivariant GNN to predict scores for the lattice, atom positions, and atom types in the denoising process. In particular, we adapt the GemNet architecture by Gastegger et al. [70], originally developed to be a universal MLFF. GemNet is a symmetric message-passing GNN that uses directional information to achieve SO(3)-equivariance, and incorporates two- and three-body information in the first layer for efficiency. Since we do not predict energies, we adapt the direct (i.e., non-conservative) force prediction variant of the model, named GemNet-dT, which has been shown to be more computationally efficient and accurate in these scenarios [70]. We employ four message-passing layers, a cutoff radius of 7 Å for the neighbor list construction, and set the dimension of hidden representations for nodes and edges to 512.

We train the model to predict Cartesian coordinate scores $\mathbf{s}_{\mathbf{X},\theta}(\mathbf{X}_t, \mathbf{L}_t, \mathbf{A}_t, t)$ as if they were non-conservative forces. These scores are equivariant to rotation and permutation, and invariant to translation. We then transform the Cartesian scores into fractional scores following Eq. (A2).

For the atom-type reverse diffusion, recall from Eq. (A16) that we predict the atom types $\mathbf{A}_0$ given the atom types $\mathbf{A}_t$ at time t . For materials, we additionally condition

on lattice $\mathbf{L}_t$ and coordinates $\mathbf{X}_t$; more precisely, the (unnormalized) log-probabilities $\log p_{\boldsymbol{\theta}}(\mathbf{A}_0|\mathbf{X}_t, \mathbf{L}_t, \mathbf{A}_t, t)$ of the atomic species at $t = 0$ are computed as:

$$\log p_{\boldsymbol{\theta}}(\mathbf{A}_0|\mathbf{X}_t, \mathbf{L}_t, \mathbf{A}_t, t) = \mathbf{H}^{(L)}\mathbf{W}, \quad (\text{A31})$$

where $\mathbf{H}^{(L)} \in \mathbb{R}^{n \times d}$ are the hidden representations of atoms at the last message-passing layer L , and $\mathbf{W} \in \mathbb{R}^{d \times K}$ are the weights of a fully-connected linear layer, with K being the number of atom types (including the masked null state).

A.8.1 Computation of the predicted lattice scores

To predict the lattice scores $\mathbf{s}_{\mathbf{L}, \boldsymbol{\theta}}(\mathbf{X}_t, \mathbf{L}_t, \mathbf{A}_t, t)$, we utilize the model’s hidden representations of the edges. For layer l , we denote the edge representation of the edge $(ijk) \in \mathcal{E}$ between atoms i and j as $\mathbf{m}_{ijk}^l \in \mathbb{R}^d$, where i is inside the unit cell and j is $\mathbf{k} \in \mathbb{Z}^3$ unit cells displaced from the center unit cell. We use a multi-layer perceptron (MLP) $\phi^l : \mathbb{R}^d \rightarrow \mathbb{R}$ to predict a scalar score per edge. We treat this as a prediction by the model indicating by how much an edge’s length should increase or decrease, and translate this into a predicted transformation of the lattice via chain rule derivation:

$$\begin{aligned} \frac{\partial \tilde{d}_{ijk}}{\partial \mathbf{L}_t} &= \frac{\partial}{\partial \mathbf{L}_t} \left\| \mathbf{L}_t \left(\mathbf{x}_t^j - \mathbf{x}_t^i + \mathbf{k} \right) \right\|_2 \\ &= \frac{1}{\tilde{d}_{ijk}} \mathbf{L}_t \left(\mathbf{x}_t^j - \mathbf{x}_t^i + \mathbf{k} \right) \cdot \left(\mathbf{x}_t^j - \mathbf{x}_t^i + \mathbf{k} \right)^\top \\ &= \frac{1}{\tilde{d}_{ijk}} \tilde{\mathbf{d}}_{ijk}(\mathbf{d}_{ijk})^\top, \end{aligned} \quad (\text{A32})$$

where $\tilde{d}_{ijk} = \|\tilde{\mathbf{d}}_{ijk}\|_2$ and $\tilde{\mathbf{d}}_{ijk} = \mathbf{L}_t \left(\mathbf{x}_t^j - \mathbf{x}_t^i + \mathbf{k} \right)$ are the edge length and edge displacement in Cartesian coordinates, respectively, and $\mathbf{d}_{ijk} = \mathbf{x}_t^j - \mathbf{x}_t^i + \mathbf{k}$ is the edge displacement in fractional coordinates. The predicted lattice score *per edge* is then $\phi^l(\mathbf{m}_{ijk}^l) \cdot \frac{\partial \tilde{d}_{ijk}}{\partial \mathbf{L}_t}$. These predicted scores are averaged over all edges $(ijk) \in \mathcal{E}$ to get the predicted lattice score for layer l :

$$\hat{\mathbf{s}}_{\mathbf{L}, \boldsymbol{\theta}}^l(\mathbf{X}_t, \mathbf{L}_t, \mathbf{A}_t, t) = \frac{1}{|\mathcal{E}|} \sum_{(ijk) \in \mathcal{E}} \phi^l(\mathbf{m}_{ijk}^l) \cdot \frac{1}{\tilde{d}_{ijk}} \tilde{\mathbf{d}}_{ijk}(\mathbf{d}_{ijk})^\top. \quad (\text{A33})$$

Stacking the model’s predictions into a diagonal matrix $\boldsymbol{\Phi}^l \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{E}|} = \text{diag} \left(\frac{\phi^l(\mathbf{m}_{ijk}^l)}{|\mathcal{E}| \cdot \tilde{d}_{ijk}} \right)$, we can write more concisely

$$\hat{\mathbf{s}}_{\mathbf{L}, \boldsymbol{\theta}}^l(\mathbf{X}_t, \mathbf{L}_t, \mathbf{A}_t, t) = \tilde{\mathbf{D}} \boldsymbol{\Phi}^l \mathbf{D}^\top = \mathbf{L}_t \mathbf{D} \boldsymbol{\Phi}^l \mathbf{D}^\top, \quad (\text{A34})$$

where $\tilde{\mathbf{D}}, \mathbf{D} \in \mathbb{R}^{3 \times |\mathcal{E}|}$ are the stacked matrices of Cartesian and fractional distance vectors, respectively, with $\tilde{\mathbf{D}} = \mathbf{L}_t \mathbf{D}$ for structure i . This form reveals that these predicted lattice scores have a key shortcoming. To see this, recall from Appendix A.7.1